Noushin Omidvar, PhD

Machine Learning Scientist | AI for Scientific Discovery

Santa Clara, California · Portfolio · LinkedIn · GitHub · Google Scholar

Profile

Machine learning scientist with a PhD in chemical engineering, working at the intersection of molecular modeling, predictive machine learning, and experimental discovery. I build models and scientific software that connect chemical data with practical decisions. My work includes molecular mixtures, model-guided screening, and interpretable models for catalysis.

Industry Experience

Machine Learning Scientist Apr 2024–Present
Aionics, Inc. Palo Alto, California

Molecular and mixture property modeling. Developed models for electrolyte viscosity, ionic conductivity, and flash point using molecular descriptors, formulation composition, and experimental conditions. Adapted graph-based deep learning for mixtures and evaluated generalization to new component combinations.

Closed-loop discovery. Built workflows that connect property prediction, chemical screening, constrained multi-objective optimization, and experimental feedback. Used NSGA-III to prioritize formulations for laboratory testing and updated selection criteria based on subsequent results.

Learning from battery experiments. Developed learning-to-rank and regression approaches for cycle-life and cell-response data to help compare formulations and interpret experimental outcomes.

Molecular screening beyond batteries. Combined chemical-space filtering, RDKit descriptors, surrogate models, and similarity-aware validation for screening candidate molecules against thermophysical-property objectives.

Scientific software and collaboration. Maintained reusable Python modules for data processing, model development, evaluation, and testing; worked with laboratory and engineering collaborators to turn model findings into candidate decisions.

Core Methods

Scientific ML: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, graph neural networks, learning-to-rank, SHAP, active learning, model validation.

Molecular discovery: RDKit, molecular descriptors, mixture modeling, screening, multi-objective optimization, DFT, VASP.

Computing: Python, NumPy, Pandas, Git, Linux, scientific software testing; working knowledge of SQL, AWS/GCP, and SLURM/HPC.

Research Experience

Graduate Research Assistant and Doctoral Researcher 2016–2022
Virginia Tech, Chemical Engineering Blacksburg, Virginia

Interpretable ML for oxygen reduction. Adapted TinNet, a theory-infused graph neural network, for OH adsorption prediction on catalyst surfaces. Modified the chemisorption module and adsorbate-orbital treatment, then trained and interpreted the model.

Electronic interpretation with SHAP. Implemented global and catalyst-specific SHAP analyses of learned electronic parameters to explain adsorption-energy trends and connect model predictions to metal–adsorbate bonding.

DFT-guided catalyst discovery. Combined density functional theory, active learning, and screening to investigate Pt-based catalyst surfaces and prioritize candidates for further study.

Collaborative electrocatalysis research. Contributed SHAP-based explanations to ammonia-oxidation research; calculated DFT reaction pathways and transition states for ethanol-oxidation research; supported other projects through adsorption modeling, DFT, figures, and writing.

Education

PhD, Chemical Engineering, Virginia Tech 2022
Dissertation: Toward Designing Active ORR Catalysts via Interpretable and Explainable Machine Learning

MS, Chemical Engineering, Tarbiat Modares University 2015
Research: Biomaterials, controlled drug delivery, and bone tissue engineering

BS, Chemical Engineering, Sharif University of Technology 2012

Selected Publications

- **Omidvar, N.** et al. "Explainable AI for optimizing oxygen reduction on Pt monolayer core–shell catalysts." *Electrochemical Science Advances* (2024). Article link.
- Pillai, H. S. et al., including **Omidvar, N.** "Interpretable design of Ir-free trimetallic electrocatalysts for ammonia oxidation with graph neural networks." *Nature Communications* (2023). Article link.
- **Omidvar, N.** et al. "Interpretable machine learning of chemical bonding at solid surfaces." *Journal of Physical Chemistry Letters* (2021). Article link.
- Zhang, Z. et al., including **Omidvar, N.** "Modular multi-interface nanocrystals for enhanced ethanol oxidation electrocatalysis." *Matter* (2026). Article link.

Additional publications: Google Scholar.